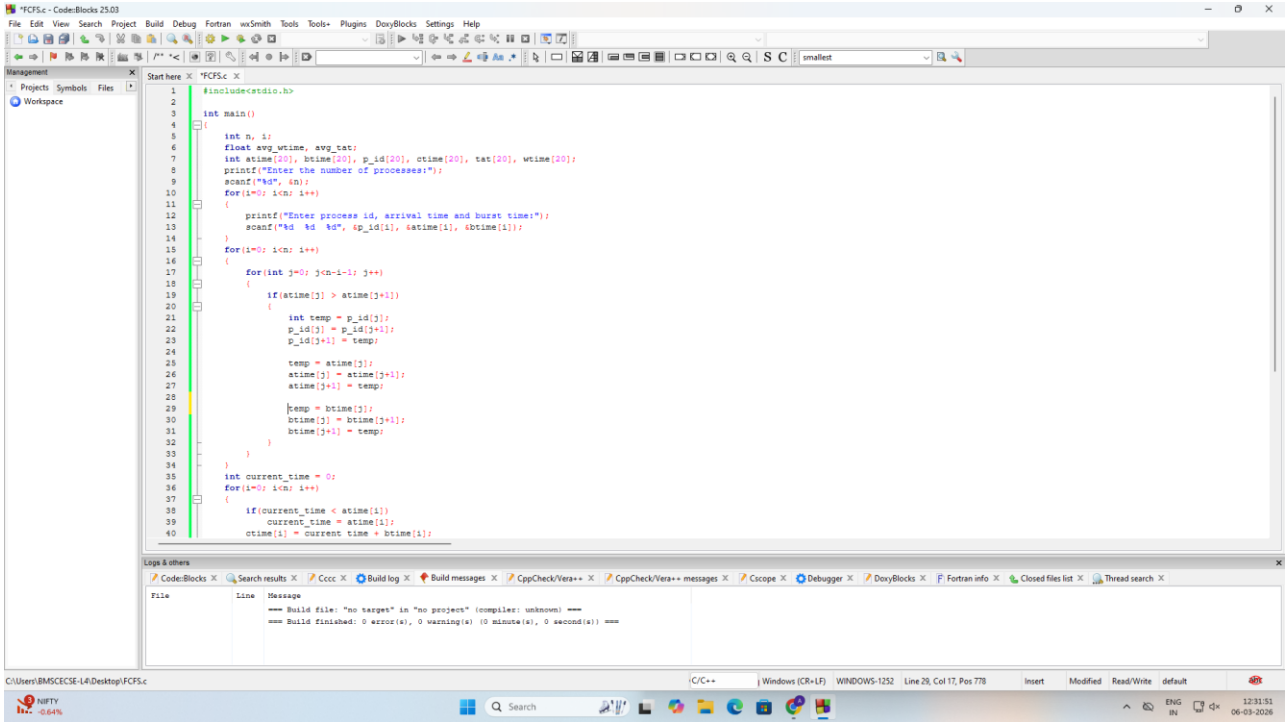
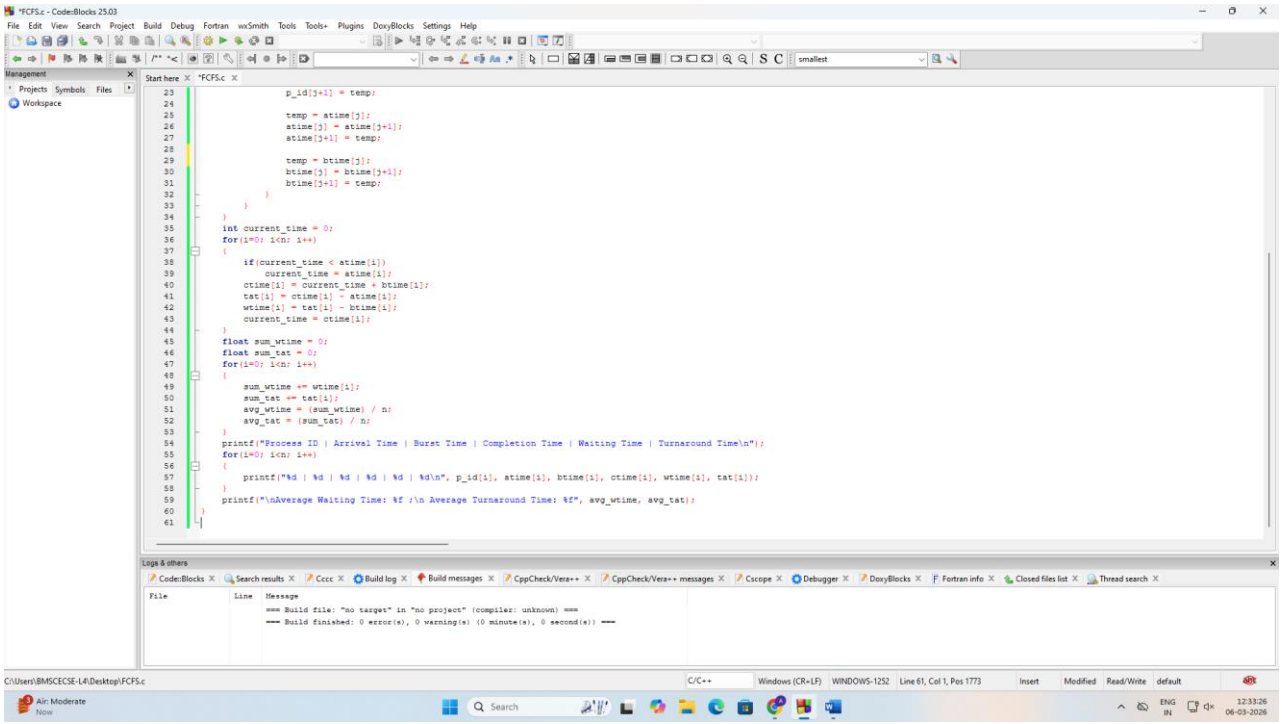


```c
#include<stdio.h>

int main()
{
    int n, i;
    float avg_wtime, avg_tat;
    int atime[20], btime[20], p_id[20], ctime[20], tat[20], wtime[20];
    printf("Enter the number of processes:");
    scanf("%d", &n);
    for(i=0; i<n; i++)
    {
        printf("Enter process id, arrival time and burst time:");
        scanf("%d  %d  %d", &p_id[i], &atime[i], &btime[i]);
    }
    for(i=0; i<n; i++)
    {
        for(int j=0; j<n-i-1; j++)
        {
            if(atime[j] > atime[j+1])
            {
                int temp = p_id[j];
                p_id[j] = p_id[j+1];
                p_id[j+1] = temp;

                temp = atime[j];
                atime[j] = atime[j+1];
                atime[j+1] = temp;

                temp = btime[j];
                btime[j] = btime[j+1];
                btime[j+1] = temp;
            }
        }
    }
    int current_time = 0;
    for(i=0; i<n; i++)
    {
        if(current_time < atime[i])
            current_time = atime[i];
        ctime[i] = current time + btime[i];
```

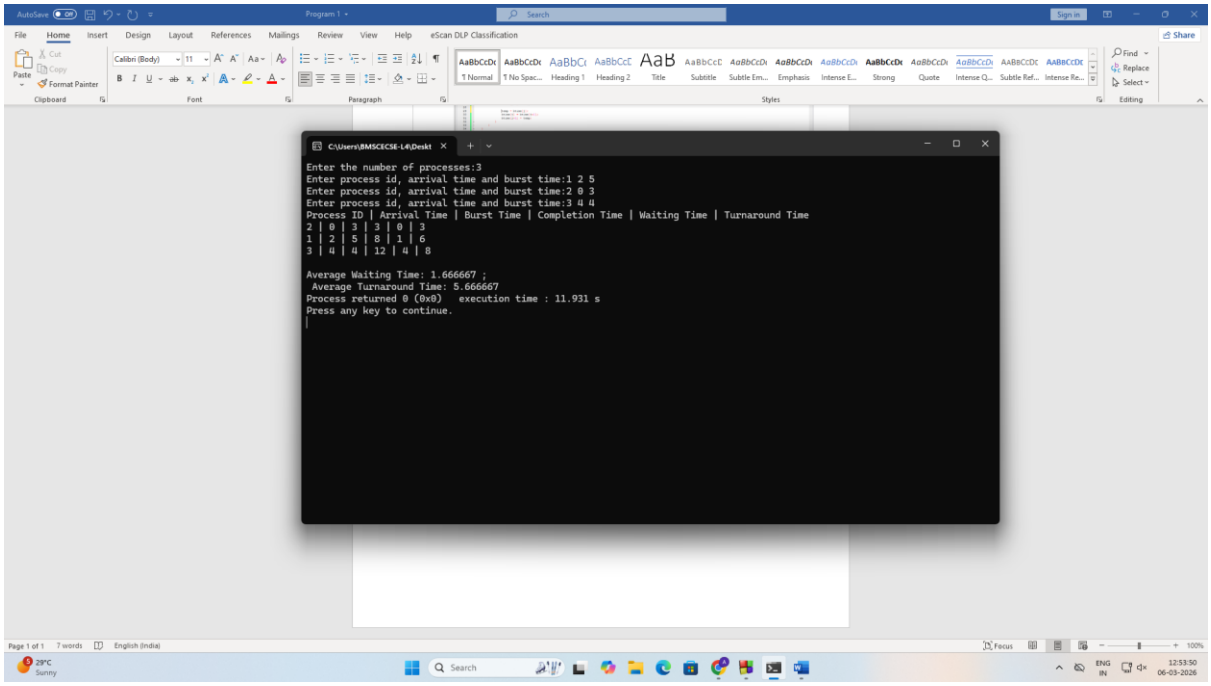


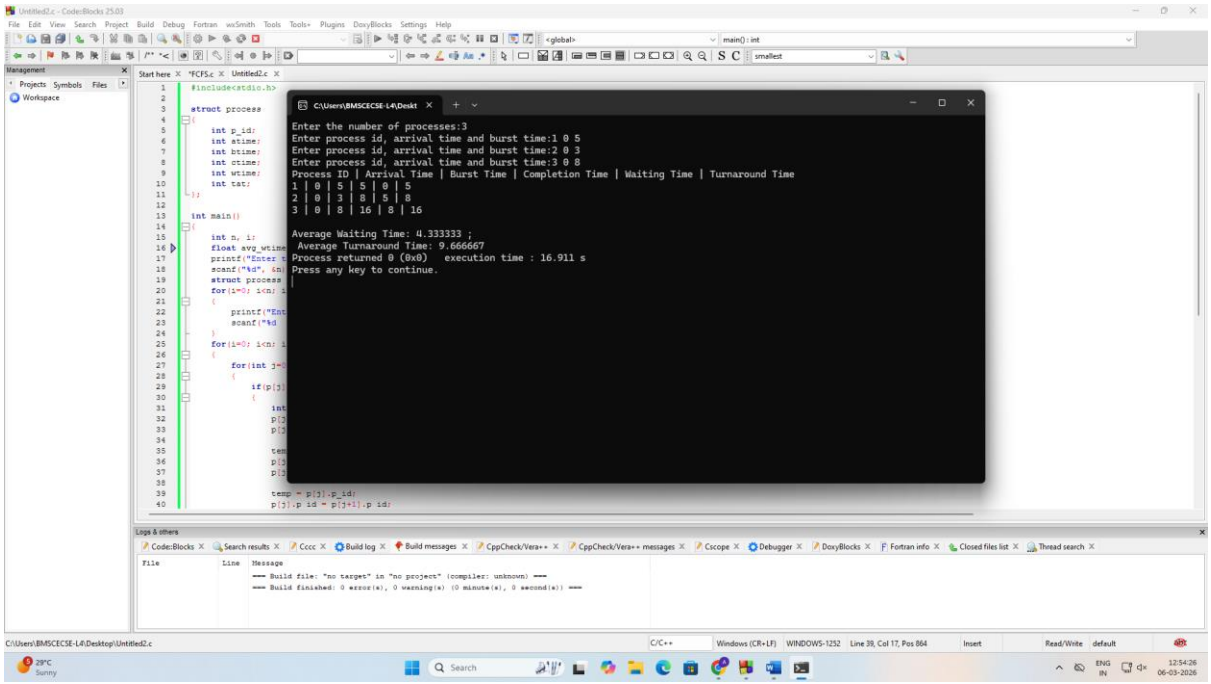
```c
                p_id[j+1] = temp;

                temp = atime[j];
                atime[j] = atime[j+1];
                atime[j+1] = temp;

                temp = btime[j];
                btime[j] = btime[j+1];
                btime[j+1] = temp;
            }
        }
    }
    int current_time = 0;
    for(i=0; i<n; i++)
    {
        if(current_time < atime[i])
            current_time = atime[i];
        ctime[i] = current_time + btime[i];
        tat[i] = ctime[i] - atime[i];
        wtime[i] = tat[i] - btime[i];
        current_time = ctime[i];
    }
    float sum_wtime = 0;
    float sum_tat = 0;
    for(i=0; i<n; i++)
    {
        sum_wtime += wtime[i];
        sum_tat += tat[i];
        avg_wtime = (sum_wtime) / n;
        avg_tat = (sum_tat) / n;
    }
    printf("Process ID | Arrival Time | Burst Time | Completion Time | Waiting Time | Turnaround Time\n");
    for(i=0; i<n; i++)
    {
        printf("%d | %d | %d | %d | %d | %d\n", p_id[i], atime[i], btime[i], ctime[i], wtime[i], tat[i]);
    }
    printf("\nAverage Waiting Time: %f ;\n Average Turnaround Time: %f", avg_wtime, avg_tat);
}
```

Program 1: First Come, First Serve (FCFS)


C:\Users\BMSCECSE-L4\Deskt
Enter the number of processes:3
Enter process id, arrival time and burst time:1 2 5
Enter process id, arrival time and burst time:2 0 3
Enter process id, arrival time and burst time:3 4 4
Process ID | Arrival Time | Burst Time | Completion Time | Waiting Time | Turnaround Time
2 | 0 | 3 | 3 | 0 | 3
1 | 2 | 5 | 8 | 1 | 6
3 | 4 | 4 | 12 | 4 | 8
Average Waiting Time: 1.666667 ;
Average Turnaround Time: 5.666667
Process returned 0 (0x0) execution time : 11.931 s
Press any key to continue.


Untitled2.c - Code::Blocks 25.03
C:\Users\BMSCECSE-L4\Deskt
Enter the number of processes:3
Enter process id, arrival time and burst time:1 0 5
Enter process id, arrival time and burst time:2 0 3
Enter process id, arrival time and burst time:3 0 8
Process ID | Arrival Time | Burst Time | Completion Time | Waiting Time | Turnaround Time
1 | 0 | 5 | 5 | 0 | 5
2 | 0 | 3 | 8 | 5 | 8
3 | 0 | 8 | 16 | 8 | 16
Average Waiting Time: 4.333333 ;
Average Turnaround Time: 9.666667
Process returned 0 (0x0) execution time : 16.911 s
Press any key to continue.

Untitled2.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here | *FCFS.c | Untitled2.c

```c
#include<stdio.h>

struct process
{
    int p_id;
    int atime;
    int btime;
    int ctime;
    int wtime;
    int tat;
};

int main()
{
    int n, i;
    float avg_wtime, avg_tat;
    printf("Enter the number of processes:");
    scanf("%d", &n);
    struct process p[n];
    for(i=0; i<n; i++)
    {
        printf("Enter process id, arrival time and burst time:");
        scanf("%d  %d  %d", &p[i].p_id, &p[i].atime, &p[i].btime);
    }
    for(i=0; i<n; i++)
    {
        for(int j=0; j<n-i-1; j++)
        {
            if(p[j].atime > p[j+1].atime)
            {
                int temp = p[j].atime;
                p[j].atime = p[j+1].atime;
                p[j+1].atime = temp;

                temp = p[j].btime;
                p[j].btime = p[j+1].btime;
                p[j+1].btime = temp;

                temp = p[j].p_id;
                p[j].p_id = p[j+1].p_id;
```

Logs & others

```
File | Line | Message
     |      | === Build file: "no target" in "no project" (compiler: unknown) ===
     |      | === Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===
```

C:\Users\BMSCECSE-L4\Desktop\Untitled2.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 39, Col 17, Pos 864 | Insert | Read/Write | default

29°C Sunny | Search | ENG IN | 12:54:54 06-03-2026

Untitled2.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here | *FCFS.c | Untitled2.c

```c
                p[j+1].atime = temp;

                temp = p[j].btime;
                p[j].btime = p[j+1].btime;
                p[j+1].btime = temp;

                temp = p[j].p_id;
                p[j].p_id = p[j+1].p_id;
                p[j+1].p_id = temp;
            }
        }
    }
    int current_time = 0;
    for(i=0; i<n; i++)
    {
        if(current_time < p[i].atime)
            current_time = p[i].atime;
        p[i].ctime = current_time + p[i].btime;
        p[i].tat = p[i].ctime - p[i].atime;
        p[i].wtime = p[i].tat - p[i].btime;
        current_time = p[i].ctime;
    }
    float sum_wtime = 0;
    float sum_tat = 0;
    for(i=0; i<n; i++)
    {
        sum_wtime += p[i].wtime;
        sum_tat += p[i].tat;
        avg_wtime = (sum_wtime) / n;
        avg_tat = (sum_tat) / n;
    }
    printf("Process ID | Arrival Time | Burst Time | Completion Time | Waiting Time | Turnaround Time\n");
    for(i=0; i<n; i++)
    {
        printf("%d | %d | %d | %d | %d | %d\n", p[i].p_id, p[i].atime, p[i].btime, p[i].ctime, p[i].wtime, p[i].tat);
    }
    printf("\nAverage Waiting Time: %f ;\n Average Turnaround Time: %f", avg_wtime, avg_tat);
}
```

Logs & others

```
File | Line | Message
     |      | === Build file: "no target" in "no project" (compiler: unknown) ===
     |      | === Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ===
```

C:\Users\BMSCECSE-L4\Desktop\Untitled2.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 39, Col 17, Pos 864 | Insert | Read/Write | default

29°C Sunny | Search | ENG IN | 12:55:22 06-03-2026